

INTERMEDIATE REPORT – **ASSIGNMENT 6**

COMPUTATIONAL BIOFLUID MECHANICS

ACADEMIC YEAR 2025-2026

Name student

Prof. Charlotte Debbaut, Prof. Patrick Segers

Dr. ir. Mohammad Rahimi Gorji, ir. Saar Vermijs, ir. Jessie Duquesne,
ir. Kaat Wille, ir. Irea Vervaeke

ASSIGNMENT 6: TRANSIENT SIMULATION OF POST-OP CASE

Make sure that figures/tables are high-quality and complete, and have numbers and clear captions.

Methods (/3.5)

- *Describe the simulation set-up (material properties, boundary conditions, etc.) and the solver settings (time step size, etc.) (/1.5)*
- *Discuss the evolution of the residuals for the transient simulation of the post-op case. How did you assess convergence? Did you reach convergence in each time step? If not, in which time steps did you not reach convergence? (/2)*

Results (/5.5)

- *Visualize the flow patterns using velocity vectors (colored according to velocity magnitude) in a figure for three relevant timepoints. Explain your choice of timepoints. (/1.5)*
- *Plot the time-varying local resistance of the same section as for the steady case (see Assignment 4) using CFD results (for a complete breathing cycle). Is this the same as you would expect for a tube of constant diameter (see Assignment 5) or not? Explain why. (/2)*
- *Make a figure where you show (A) the flow rate of the breathing profile you used as an inlet boundary condition, and sketch (B) the corresponding time-varying local resistance for the pre-op and post-op case (qualitatively!). Can you explain the difference in behavior for the resistance between the pre-op and post-op case? (/2)*

Discussion (/1)

If you did not reach convergence in each time step, how would you tackle this problem give more time and computational resources? (/1)